EE6094
CAD for VLSI Design

Compiler and Checker

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Outline

- Compiler
- Checker

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Compiler

- How to compile and execute your source code on workstation?
  - Step1: Use g++ command

```
[[redacted]@cad ~/PA1]$ g++ -std=c++11 test.cpp -o test
```

  - Step2: Execute your output file

```
[[redacted]@cad ~/PA1]$ ./test
Hello PA1!
```

./Your output file name

Outline

- Compiler
- Checker

Checker

- How to check the Verilog code dumped by your C++ code is correct?
  - We need to use testbench to check the result.

Checker

- How to use the checker?
  - Step1: Generate an executable file (ex. exe)
  - Step2: Make sure you put netlist files, testbench code files, Makefile and executable file in the same folder

```
[[redacted]@ncuee PA1]$ g++ -std=c++11 PA1.cpp -o exe
```

```
[[redacted]@ncuee PA1]$ ls
c17.netlist  c17_testbench.v  exe  PA1.cpp
```


Checker

- How to use the checker?
  - Step3: Key in the following commands
    source /usr/cad/cadence/CIC/incisiv.cshrc
    source /usr/cad/synopsys/CIC/verdi.cshrc

```
[[redacted]@ncuee PA1]$ source /usr/cad/cadence/CIC/incisiv.cshrc
set INCISIV version: INCISIVE_15.20.084 (default)
[[redacted]@ncuee PA1]$ source /usr/cad/synopsys/CIC/verdi.cshrc
set verdi version: 2019.06/ (default)
```

  - Step4: Use your executable file to translate the netlist file to Verilog format.

```
[[redacted]@ncuee PA1]$ ./exe c17.netlist c17.v
[[redacted]@ncuee PA1]$ ls
c17.netlist  c17_testbench.v  c17.v  exe  PA1.cpp
```

Checker

- How to use the checker?
  - Step5: Check correctness of your result.

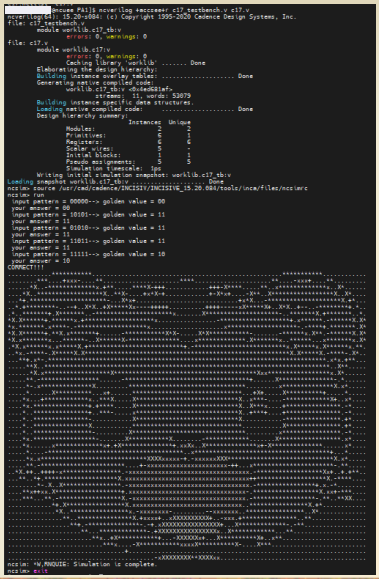
```
[[redacted]ncuee PA1]$ ncverilog +access+r c17_testbench.v c17.v
```

  - Step6: If you want to use check again, you need to remove some file which are generated by checker last time.

```
[[redacted]ncuee PA1]$ ls
c17.netlist  c17_testbench.v  c17.v  exe  INCA_libs  ncverilog.history  ncverilog.log  PA1.cpp
[[redacted]ncuee PA1]$ rm -r INCA_libs/
[[redacted]ncuee PA1]$ ls
c17.netlist  c17_testbench.v  c17.v  exe  ncverilog.history  ncverilog.log  PA1.cpp
```


Checker

- Correct result



Checker

- Wrong result

```
[oem P413] ncverilog +access+r c17_testbench.v c17.v
ncvlog(64): 15.20-s004: (c) Copyright 1995-2020 Cadence Design Systems, Inc.
Recompiling.. /ncsim: file './c17.v' is newer than expected.
        expected: Thu Feb 27 10:45:25 2025
        actual:   Thu Feb 27 10:47:43 2025
file: c17.v
        module worklib.c17:v
                errors: 0, warnings: 0
                Caching library 'worklib' ........ Done
        Elaborating the design hierarchy:
        Building instance overlay tables: ........................ Done
        Generating native compiled code:  ........................ Done
        Loading native compiled code:     ........................ Done
        Design hierarchy summary:  Instances  Unique
                Modules:                   2       2
                Primitives:                6       2
                Registers:                 6       6
                Scalar wires:              5       -
                Initial blocks:            1       1
                Pseudo assignments:        5       5
                Simulation timescale:  1ps
        Writing initial simulation snapshot: worklib.c17_sb:v
Loading snapshot worklib.c17_sb:v ........................ Done
ncsim: source /usr/cad/cadence/INCISIV/INCISIVE_15.20.004/tools/inca/files/ncsim.rc
ncsim> run
  input pattern = 00000--> golden value = 00
  your answer = 10
  input pattern = 10101--> golden value = 11
  your answer = 01
  input pattern = 01010--> golden value = 11
  your answer = 11
  input pattern = 11011--> golden value = 11
  your answer = 11
  input pattern = 11111--> golden value = 10
  your answer = 00
WRONG!!!!!
ncsim: *W,MODEXI: Simulation is complete.
ncsim> exit
```